

## g2(0) cases

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abstract

We consider a negatively charged quantum dot (QD) with a resident electron embedded in a symmetric optical microcavity. The optically active transition couples the ground state, in which the QD hosts a single resident electron, to the negatively charged trion  $X^-$  with energy  $E_t$ , built of an electron singlet and a heavy hole. The energy of the fundamental cavity mode  $\hbar\omega_0$  coincides with that of the trion transition,  $E_t = \hbar\omega_0$ , and the interaction with all other modes is neglected. The system is driven by a laser pulse of carrier frequency  $\omega_L$ , detuned from the transition by  $\Delta = \omega_L - \omega_0$ , and is subject to a magnetic field  $B_x$ , perpendicular to the growth axis. We are interested in the non-resonant regime, in which the detuning exceeds both the spectral width  $\Gamma_L$  of the pump pulse and the width  $\Gamma$  of the cavity mode,  $\omega_L - \omega_0 \gtrsim \Gamma_L, \Gamma$ , as sketched in Fig. 1. As the pump pulse carries a large number of photons, this separation of scales allows to describe it as a classical field with the Rabi frequency  $\Omega_R(t)$ , whereas the cavity mode, which is not directly fed by the laser, contains only a few photons and therefore has to be retained at the operator level.

Within the rotating wave approximation the Hamiltonian of the system, which accounts for the coupling to the cavity mode, for the interaction with the driving laser field and for the interaction with the magnetic field, reads (hereafter  $\hbar = 1$ )

$$\begin{aligned} \hat{H} = & E_t \sum_{\sigma=\uparrow,\downarrow} \hat{d}_\sigma^\dagger \hat{d}_\sigma + \omega_0 \sum_{\lambda=\pm} \hat{b}_\lambda^\dagger \hat{b}_\lambda \\ & + g \left( \hat{d}_\uparrow^\dagger \hat{a}_\uparrow \hat{b}_+ + \hat{d}_\downarrow^\dagger \hat{a}_\downarrow \hat{b}_- \right) + g^* \left( \hat{d}_\uparrow \hat{a}_\uparrow^\dagger \hat{b}_+^\dagger + \hat{d}_\downarrow \hat{a}_\downarrow^\dagger \hat{b}_-^\dagger \right) \\ & + \Omega_R(t) \sum_{\sigma=\uparrow,\downarrow} \left( e^{-i\omega_L t} \hat{d}_\sigma^\dagger \hat{a}_\sigma + e^{i\omega_L t} \hat{d}_\sigma \hat{a}_\sigma^\dagger \right) \\ & + \mu_B g_e B_x \left( \hat{a}_\downarrow^\dagger \hat{a}_\uparrow + \hat{a}_\uparrow^\dagger \hat{a}_\downarrow \right) + \mu_B g_t B_x \left( \hat{d}_\downarrow^\dagger \hat{d}_\uparrow + \hat{d}_\uparrow^\dagger \hat{d}_\downarrow \right) \end{aligned} \quad (1)$$

where the last two terms describe the Zeeman interaction in the transverse field. Here  $\hat{a}_\sigma^\dagger$  is the creation operator of the resident electron, where  $\sigma = \uparrow, \downarrow$  denotes the spin state  $|\pm 1/2\rangle$ ;  $\hat{d}_\sigma^\dagger$  is the creation operator of the negatively charged trion, where  $\sigma$  denotes the heavy hole spin state  $|\pm 3/2\rangle$ ;  $\hat{b}_\pm^\dagger = (\hat{b}_H^\dagger \pm i\hat{b}_V^\dagger)/\sqrt{2}$  is the creation operator of a cavity photon with circular polarization  $\sigma_\pm$ , expressed through the linearly polarized modes  $H$  and  $V$ ;  $g$  is the effective QD-cavity coupling parameter;  $\mu_B$  is the Bohr

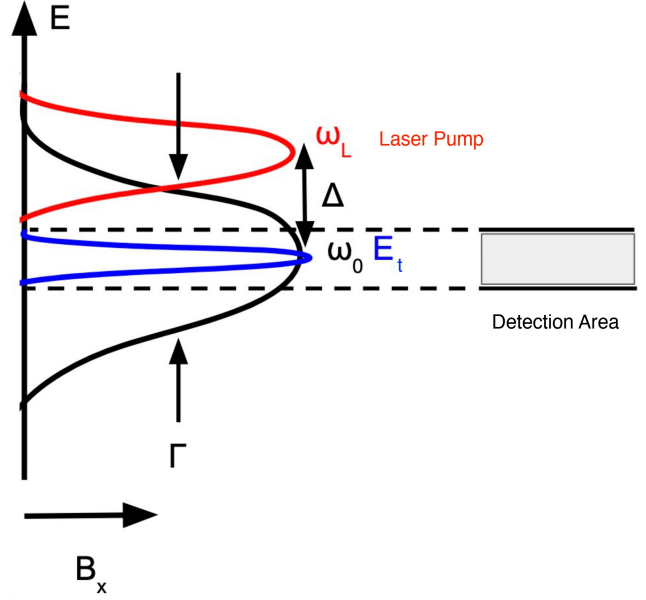


FIG. 1. Energy scheme of the system. The cavity mode of width  $\Gamma$  (black) is centered at  $\omega_0$  and is resonant with the trion transition  $E_t$  (blue). The pump pulse (red) has the carrier frequency  $\omega_L$  and the spectral width  $\Gamma_L$ , and is detuned from the mode by  $\Delta$ . The shaded box marks the spectral window of detection. The magnetic field  $B_x$  is applied perpendicular to the growth axis.

magneton,  $g_e$  and  $g_t$  are the in-plane  $g$ -factors of the resident electron and of the trion, respectively.

The optical selection rules fix the spin structure of the light-matter coupling: a  $\sigma_+$  photon converts a resident electron  $|+1/2\rangle$  into the trion with the heavy hole  $|+3/2\rangle$ , and a  $\sigma_-$  photon converts  $|-1/2\rangle$  into  $|-3/2\rangle$ . Consequently, only the triads  $(\hat{a}_\uparrow, \hat{d}_\uparrow, \hat{b}_+)$  and  $(\hat{a}_\downarrow, \hat{d}_\downarrow, \hat{b}_-)$  are coupled, and the polarization of the emitted photon is rigidly tied to the spin state of the QD. The transverse field  $B_x$ , in turn, mixes the two spin branches, which is the mechanism that correlates the polarizations of the two emitted photons.

To describe the evolution of the system coupled to its environment, the Hamiltonian dynamics has to be supplemented by the dissipative terms of the Lindblad equation

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$$\partial_t \hat{\rho} = -i [\hat{H}, \hat{\rho}] + \sum_i \gamma_i \left( \hat{V}_i \hat{\rho} \hat{V}_i^\dagger - \frac{1}{2} \{ \hat{V}_i^\dagger \hat{V}_i, \hat{\rho} \} \right) \equiv \hat{\mathcal{L}} \hat{\rho} \quad (2)$$

where  $\{\cdot, \cdot\}$  is the anticommutator, and  $\gamma_i$  and  $\hat{V}_i$  are the rate and the jump operator of the  $i$ -th channel. Three dissipation channels are retained.

The first one is the escape of photons from the cavity mode into the environment, which is the process that delivers the photons to the detector. Its jump operator and rate are

$$\hat{V}_{\text{phot}} = \hat{b}_\pm, \quad \gamma_{\text{phot}} = \Gamma \quad (3)$$

where  $\Gamma$  is the width of the cavity mode.

The second channel is the pure dephasing of the optical transition induced by acoustic phonons, which is responsible for the damping of the Rabi oscillations. The corresponding jump operator is

$$\hat{V}_{\text{phon}} = \sum_{\sigma=\uparrow, \downarrow} \left( \hat{a}_\sigma^\dagger \hat{a}_\sigma - \hat{d}_\sigma^\dagger \hat{d}_\sigma \right), \quad \gamma_{\text{phon}} = \alpha \Omega_R^2 \quad (4)$$

The third channel is the dephasing of the resident electron spin. It is described by the operator

$$\hat{V}_{\text{spin } i} = \begin{pmatrix} \hat{a}_\uparrow^\dagger & \hat{a}_\downarrow^\dagger \end{pmatrix} \hat{\sigma}_i \begin{pmatrix} \hat{a}_\uparrow \\ \hat{a}_\downarrow \end{pmatrix} + \begin{pmatrix} \hat{d}_\uparrow^\dagger & \hat{d}_\downarrow^\dagger \end{pmatrix} \hat{\sigma}_i \begin{pmatrix} \hat{d}_\uparrow \\ \hat{d}_\downarrow \end{pmatrix} \quad (5)$$

where  $\hat{\sigma}_i$  with  $i = x, y, z$  are the Pauli matrices acting in the spin space of the resident electron and of the trion. Since the magnetic field is directed along  $x$ , the operators  $\hat{V}_{\text{spin } y}$  and  $\hat{V}_{\text{spin } z}$  relax the longitudinal spin component, whereas  $\hat{V}_{\text{spin } x}$  adds a pure dephasing of the transverse ones. The three rates are therefore expressed through the longitudinal and transverse spin relaxation times  $T_1$  and  $T_2$  as

$$\gamma_{\text{spin } x} = \frac{1}{T_1}, \quad \gamma_{\text{spin } y} = \gamma_{\text{spin } z} = \frac{1}{T_2} \quad (6)$$

This channel destroys the coherence between the two spin branches and therefore directly degrades the polarization entanglement of the emitted photon pair.

Equation (2) is solved with the initial condition set by the state of the system before the arrival of the pump pulse, when the cavity mode is empty and the QD hosts only the resident electron. In the transverse magnetic field the eigenstates of the electron spin are the symmetric and antisymmetric superpositions  $(|\uparrow\rangle \pm |\downarrow\rangle)/\sqrt{2}$  split by the energy  $2\mu_B g_e B_x$ , and their populations follow the Boltzmann distribution at the lattice temperature  $T$ . The initial density matrix therefore reads

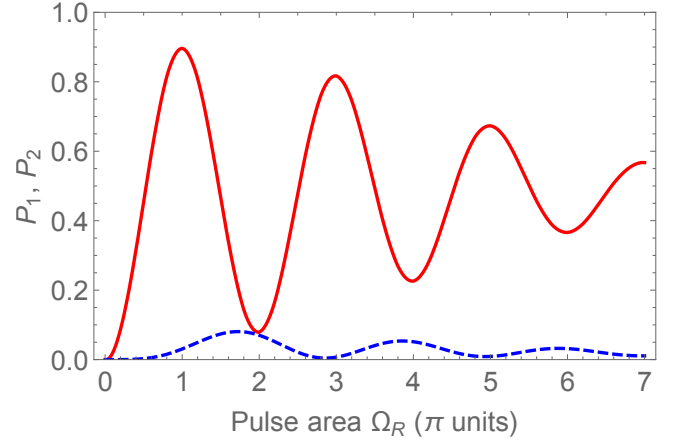


FIG. 2. Probabilities of the one-photon,  $P_1$  (solid), and two-photon,  $P_2$  (dashed), emission as functions of the pump pulse area.

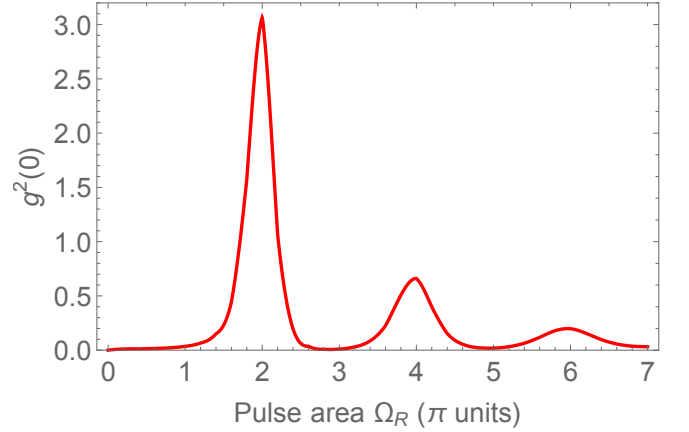


FIG. 3. Second-order correlation function at zero delay,  $g^{(2)}(0)$ , as a function of the pump pulse area.

$$\hat{\rho}_0 = \frac{f(B_x)}{2} \left( \hat{a}_\uparrow^\dagger + \hat{a}_\downarrow^\dagger \right) |0\rangle \langle 0| \left( \hat{a}_\uparrow + \hat{a}_\downarrow \right) + \frac{f(-B_x)}{2} \left( \hat{a}_\uparrow^\dagger - \hat{a}_\downarrow^\dagger \right) |0\rangle \langle 0| \left( \hat{a}_\uparrow - \hat{a}_\downarrow \right) \quad (7)$$

where  $f(B_x) = e^{\mu_B g_e B_x / T} / (e^{\mu_B g_e B_x / T} + e^{-\mu_B g_e B_x / T})$  and the temperature is measured in energy units.

All the results presented below are obtained by a numerical solution of Eq. (2) with the following set of parameters: the cavity mode width  $\Gamma = 50 \mu\text{eV}$ , the QD-cavity coupling  $g = 25 \mu\text{eV}$ , the detuning  $\Delta = 50 \mu\text{eV}$ , the pulse duration  $\tau = 13 \text{ ps}$ , the phonon coupling constant  $\alpha = 0.2 \text{ ps}$ , the transverse magnetic field  $B_x = 5 \text{ T}$  and the lattice temperature  $T = 1 \text{ K}$ . The pulse area  $\theta = \int \Omega_R(t) dt$  is varied by changing the pump power at a fixed pulse duration, and is measured below in units of  $\pi$ .

The emission probabilities are shown in Fig. 2. The one-photon probability  $P_1$  exhibits damped Rabi oscilla-

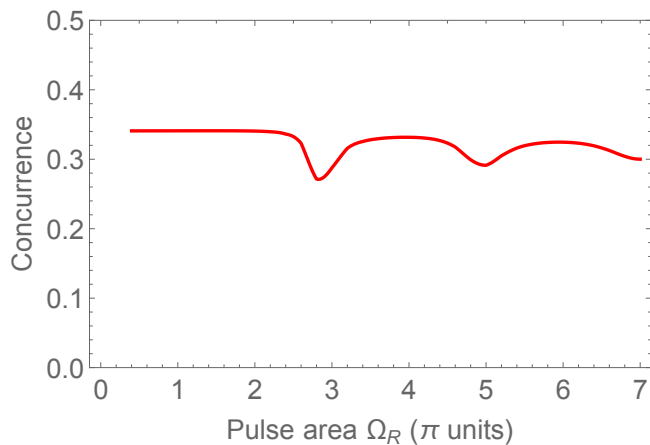


FIG. 4. Concurrence of the emitted two-photon polarization state as a function of the pump pulse area.

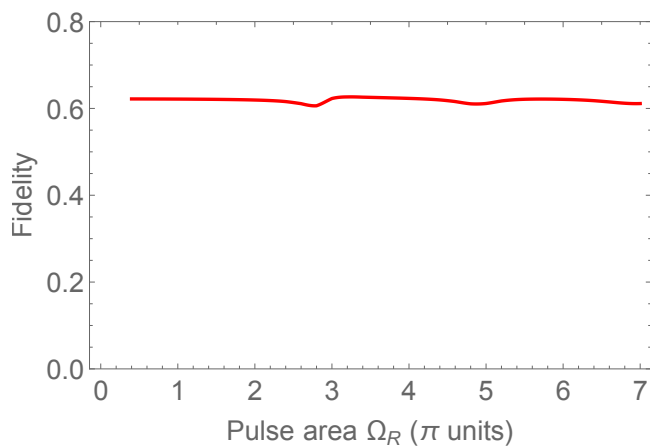


FIG. 5. Fidelity of the emitted two-photon polarization state to the Bell state  $|\Phi^+\rangle$  as a function of the pump pulse area.

tions with maxima at odd and minima at even multiples of  $\pi$ , the damping being caused by the phonon-induced dephasing. The two-photon probability  $P_2$  oscillates in antiphase and remains an order of magnitude smaller, reaching its maxima close to even pulse areas.

This antiphase behaviour is directly reflected in the second-order correlation function, Fig. 3. At odd pulse areas  $g^{(2)}(0)$  nearly vanishes, so that the source emits single photons, whereas at even ones it displays pronounced bunching peaks, the largest reaching  $g^{(2)}(0) \approx 3$  at  $\theta = 2\pi$ .

The polarization state of the emitted pair is characterized in Figs. 4 and 5. Both the concurrence and the fidelity to the Bell state  $|\Phi^+\rangle$  are only weakly sensitive to the pump power, staying close to 0.34 and 0.62, respectively, over the whole range of pulse areas, with shallow minima near the odd ones. The fidelity exceeding  $1/2$  confirms that the emitted pair is entangled for any excitation power.